

Lab 1.

No submission for this lab. The solutions will be published on Moodle right after we are finished with the lab (each lab spans two weeks usually).

Exercise 1: Agent Performance and Utility. ()

Consider both the explanations of performance measure and utility function from the lecture slides. Then, answer the following two questions.

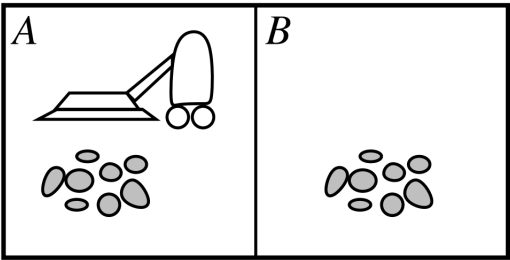
- (i) What is the difference between a performance measure and a utility function?
- (ii) Describe the relation between the performance measure (Critic) and the utility function (Performance element) for a learning agent.

Exercise 2: Agent Perception. ()

Review the table-driven (fig. 1) vacuum cleaning agent and its agent table (fig. 2). Let P be the set of possible percepts and T the lifetime of the agent (the total number of percepts it will receive).

- (i) What is the size of P for the vacuum cleaning agent?
- (ii) How many entries does the lookup table of the vacuum cleaning agent contain after 3 time steps?
 - (a) if we store only the percepts that were perceived?
 - (b) if we store all possible percepts?
- (iii) How many entries will the lookup table contain if we store all possible percepts, i.e. all possible percept sequences for the whole life time of the agent? Please give the general formula in respect to the lifetime T .

Vacuum-cleaner world



Percepts: location and contents, e.g., $[A, Dirty]$

Actions: *Left, Right, Suck, NoOp*

Figure 1: Ex 2. Vacuum cleaning agent

Percept sequence	Action
$[A, Clean]$	<i>Right</i>
$[A, Dirty]$	<i>Suck</i>
$[B, Clean]$	<i>Left</i>
$[B, Dirty]$	<i>Suck</i>
$[A, Clean], [A, Clean]$	<i>Right</i>
$[A, Clean], [A, Dirty]$	<i>Suck</i>
$\vdots$	$\vdots$

Figure 2: Ex.2 Lookup table of vacuum cleaner agent

Exercise 3: Agent Rationality.

()

Again consider the example of the vacuum cleaner agent with the following specifications:

- The "geography" is known a priori, but not the distribution of dirt and the location of the agent.
- Clean squares stay clean, sucking removes dirt.
- Left and right moves move the agent to the other location, unless they would take the agent outside the room. In this latter situation, the agent stays where it is.
- The only available actions are LEFT, RIGHT, SUCK.
- The agent correctly perceives its location and whether there is dirt.
- The performance measure awards 1 point for each new clean square at each time step.

function REFLEX-VACUUM-AGENT(*[location, status]*) **returns** an action

if *status* = *Dirty* **then return** *Suck*

else if *location* = *A* **then return** *Right*

else if *location* = *B* **then return** *Left*

Figure 3: Action function of vacuum cleaner agent

Given the information above, is the simple vacuum cleaner function in Figure 3 rational or not? Please justify your answer by 2-3 sentences. (Argumentations should be based on post-handout slide 9 definitions.)

Exercise 4: State Space.**()**

Consider the state space depicted in Figure 4, where A is the initial state, and F and I are goal states. The transitions are annotated by their costs. List all states that are

- (i) solvable,
- (ii) dead-ends,
- (iii) not reachable from G ,
- (iv) reachable from H .

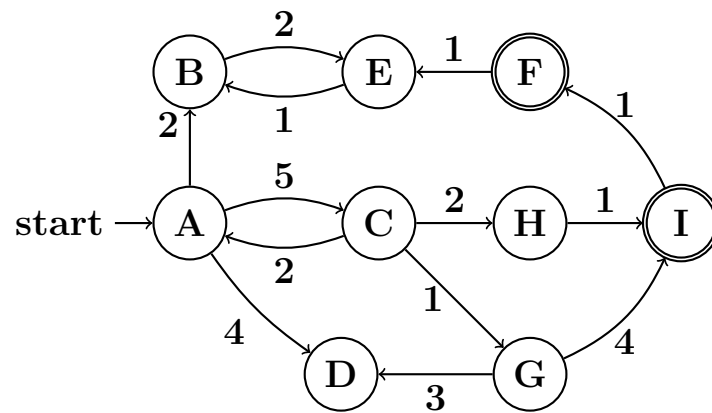


Figure 4: State space used in Exercise 4.